Quiz 6: Metals

Review of Session 6. ~15 minutes. Calculator needed.

Section A: Identifying the six metals

1. Which metal on the list is reddish-orange?
2. Which metal on the list is magnetic?
3. Which two metals are much lighter (lower density) than the rest?
4. Which metal does **not** react with hydrochloric acid at all?
5. Give the atomic number: Mg, Al, Fe, Cu, Zn.

Section B: Density

1. Density = ______ ÷ ______.
2. What is the density of water in g/mL?
3. Convert 2.70 g/cm^3 to kg/m^3.
4. A metal cube 2.0 cm on a side has a mass of 21.6 g. What is its density? Which metal is it?
5. A chunk of metal raises water from 15.0 mL to 24.0 mL and weighs 70.7 g. Density? Metal?

Section C: Reactions, oxides, alloys

1. Name the four basic reaction types.
2. A metal dissolving in acid ($Zn + 2\ HCl \rightarrow ZnCl_2 + H_2$) is which type?
3. Why does aluminum resist acid at first?
4. $2\ Mg + O_2 \rightarrow 2\ MgO$ is which reaction type?
5. Name the metals in: steel, brass, bronze.

Answer Key

Section A

1. Copper.
2. Iron.
3. Magnesium and aluminum.
4. Copper.
5. Mg 12, Al 13, Fe 26, Cu 29, Zn 30.

Section B

1. mass ÷ volume.
2. 1.00 g/mL.
3. 2700 kg/m^3.
4. Volume = 8.0 cm^3; 21.6 ÷ 8.0 = 2.70 g/cm^3 → aluminum.
5. Volume = 9.0 mL; 70.7 ÷ 9.0 ≈ 7.86 g/cm^3 → iron.

Section C

1. Synthesis, decomposition, single replacement, double replacement.
2. Single replacement.
3. A protective aluminum-oxide (Al_2O_3) passivation layer forms in air; the acid must dissolve it before reaching the metal.
4. Synthesis (combination).
5. Steel = iron + carbon; brass = copper + zinc; bronze = copper + tin.